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The Impacts of Tropospheric Gravity Wave-Generated MSTIDs on Skywaves at Mid-latitude North American Sector Observed and Modeled Using SuperDARN HF Radars

S. Chakraborty[1], P. A. Inchin[2], S. Debchoudhury[1], B. Bergsson[1], M. Zettergren[1], and J. M. Ruohoniemi[3]

[1]Embry-Riddle Aeronautical University, Daytona Beach, FL, USA, [2]Computational Physics Inc., Springfield, VA, USA, [3]Center for Space Science and Engineering Research, Blacksburg, VA, USA

Key Points:

- Thunderstorm-generated medium-scale traveling ionospheric disturbances (MSTIDs) modulate HF propagation causing oscillations in SuperDARN ground/sea scatter skip distance
- Coupled MAGIC–GEMINI simulations reproduce key MSTID-driven HF propagation signatures observed by SuperDARN and GNSS Total Electron Content observations

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

S. Chakraborty,
chakras4@erau.edu

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Author Contributions:

Conceptualization: S. Chakraborty, S. Debchoudhury
Data curation: P. A. Inchin
Formal analysis: S. Chakraborty, S. Debchoudhury, B. Bergsson, J. M. Ruohoniemi
Funding acquisition: S. Debchoudhury
Investigation: S. Chakraborty, B. Bergsson
Methodology: S. Chakraborty, P. A. Inchin

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Abstract Trans-ionospheric high frequency (HF: 3–30 MHz) response to atmospheric gravity waves (GWs) is studied in the middle-latitude ionosphere in relation to thunderstorm activity. SuperDARN HF radar observations are compared against the model simulations to quantify the impact of GW-generated medium-scale traveling ionospheric disturbances (MSTID) activity on the skywaves traveling through ionospheric F-region heights. The tropospheric thunderstorm-driven convective source is modeled using MAGIC. The outputs are coupled with GEMINI to model ionospheric plasma response, which is then used to model SuperDARN HF radar observations using the PHaRLAP raytracing tool. Semi-concentric GWs were observed at different atmospheric heights, creating medium-scale traveling ionospheric disturbances (MSTIDs) at F-region heights. PHaRLAP raytracing through the modeled ionosphere shows qualitative agreement with SuperDARN daytime ground/sea scatter observations. Modeled rays show possibilities of long ducting Pedersen rays, suggesting MSTID can create a plasma waveguide to duct rays at the F-region altitudes.

Plain Language Summary Thunderstorms do more than produce rain, lightning, and strong winds, they can also send atmospheric waves upward into space. These waves, called atmospheric gravity waves, travel from the lower atmosphere into the ionosphere, a region of charged particles about 100–400 km above Earth. When these waves reach the ionosphere, they can disturb it and create ripple-like structures known as medium-scale traveling ionospheric disturbances (MSTIDs). In this study, we use SuperDARN radar observations and GPS-based measurements to examine how these disturbances affect high-frequency (HF) radio waves used for long-distance communication. We also use computer models to simulate how thunderstorms generate upward waves, how the ionosphere responds, and how radio waves travel through the disturbed region. Our results show that thunderstorm-driven atmospheric gravity waves can significantly modify the ionosphere and even guide radio waves over long distances. The simulations agree well with observations, highlighting how weather in the lower atmosphere can influence radio systems in near-Earth space.

1. Introduction

Atmospheric gravity waves (GWs) are mechanical oscillations driven by buoyancy restoring forces acting against gravity (e.g., Francis, 1973; Godin, 2012; Hines, 1960; Yeh & Liu, 1974), stemming from various sources, including convective storms, jet stream instabilities, topographic forcing, natural hazards (e.g., tsunamis or volcanic eruptions) and human activity (e.g., Azeem et al., 2015; Hickey et al., 2009; Nishioka et al., 2013; Plougonven & Zhang, 2014). As GWs propagate vertically, their behavior is modulated by atmospheric stratification and background winds (Fritts & Alexander, 2003; Kshevetskii et al., 2021). These vertically propagating GWs often induce perturbations in the ionosphere (with wavelengths ranging between ~80–500 km), known as traveling ionospheric disturbances (TIDs). To distinguish from other TID sources, we refer to these as GW-TIDs, which are characterized by periods of 13 to ~60 min, wavelengths spanning a few tens of kilometers to a few hundreds of kilometers, and horizontal propagation speeds ranging from 100 to 250 m/s (e.g., Ogawa et al., 1987). These perturbations result from collisions between neutrals and plasma, with dynamics further shaped by geomagnetic and electrodynamic influences (Hickey et al., 2009; Hines, 1960). In the lower ionosphere (D- and E-regions), GW-TIDs predominantly follow neutral motions, whereas at F-region altitudes, plasma responses are more aligned with magnetic field lines (Hooke, 1969; Schunk & Nagy, 2000).

Project administration: S. Debchoudhury
Resources: P. A. Inchin
Software: S. Chakraborty, P. A. Inchin, B. Bergsson, M. Zettergren
Validation: S. Chakraborty, S. Debchoudhury, J. M. Ruohoniemi
Visualization: S. Chakraborty, P. A. Inchin, S. Debchoudhury
Writing – original draft: S. Chakraborty
Writing – review & editing: P. A. Inchin, S. Debchoudhury, B. Bergsson, M. Zettergren, J. M. Ruohoniemi

Traveling ionospheric disturbances (TIDs) are wave-like perturbations in ionospheric plasma density that are often generated by atmospheric GWs propagating upward from lower atmospheric or auroral sources (Hines, 1960; Hooke, 1968). TIDs are commonly categorized into large-scale TIDs (LSTIDs) and medium-scale TIDs (MSTIDs) based on their horizontal wavelengths and temporal scales (Hunsucker, 1982). LSTIDs typically have horizontal wavelengths exceeding approximately 1,000 km and periods of several hours and are often associated with geomagnetic storm-time energy input at high latitudes (Hayashi et al., 2010; Tsugawa & Saito, 2004). In contrast, MSTIDs exhibit horizontal wavelengths of a few hundred kilometers and periods of tens of minutes, and they are frequently generated by atmospheric GWs originating from sources such as tropospheric convection, meteorological fronts, or other lower-atmospheric disturbances.

MSTIDs have been linked to both electrodynamic processes (e.g., Lin et al., 2022; Shinbori et al., 2022; Zhang et al., 2022) and atmospheric GWs. MSTIDs driven by electrodynamic enhancements typically occur at night and are observed as airglow brightenings. These are often associated with increased plasma–neutral coupling under specific geometrical conditions (Kelley, 2011; Ogawa et al., 2009), which are beyond the scope of this study. MSTID impacts have been examined using both radio-based instruments (e.g., radar observations reported by Samson et al., 1989; Samson et al., 1990; Frissell et al., 2014), ionosondes (Fu et al., 2025), and GPS Total Electron Content measurements (Komjathy et al., 2016), as well as optical techniques such as all-sky imagers and Fabry–Perot interferometers (Ogawa et al., 2009).

In a vertically stratified ionosphere, high-frequency (HF) rays can travel long distances—up to 5,000 km—via ducting, a mechanism commonly referred to as the Pedersen mode (e.g., Erukhimov et al., 1997). Pedersen rays are characterized by their divergent paths that glide along the peak of the F-region ionosphere (hmF_2) (Ponomarenko & McWilliams, 2023). In this context, "divergent" refers to the tendency of Pedersen ray paths to spread apart as they glide along the F-region peak (hmF_2), rather than remaining parallel. This spreading arises from vertical and horizontal electron density gradients in the ionosphere, which refract nearby rays differently. As a result, the wave energy becomes distributed over a wider area, often reducing signal strength compared to more confined propagation modes. Pedersen rays exhibit distinct propagation characteristics, which depend on the HF operating frequency, ray elevation angle, and ground range (Davies, 1990). Due to their exponential divergence with distance, Pedersen rays are typically observed within a narrow elevation angle range at moderately high elevation angles under quiet ionospheric conditions (Ponomarenko et al., 2015). Long-distance HF propagation in the Pedersen mode is thought to be supported by structured ionospheric features that confine the rays near the F-layer maximum. These guiding structures can result from GW-induced stratification (M. Tinin, 1983) or from large-scale, field-aligned irregularities formed along the layer (M. V. Tinin et al., 1992). Previous studies have combined SuperDARN observations with gravity-wave modeling to investigate TIDs. For example (Hayashi et al., 2010), compared SuperDARN radar and GNSS TEC observations with numerical gravity-wave analyses and showed that the observed LSTIDs were consistent with gravity-wave propagation.

This study focuses on MSTIDs generated by convective sources, specifically thunderstorms, and investigates how these structures modify HF radio-wave propagation. Using a coupled first-principles modeling framework (MAGIC + GEMINI) together with PHaRLAP ray tracing, we examine how MSTID-induced ionospheric perturbations can reorganize HF propagation paths and enhance specific propagation modes such as Pedersen rays. This approach enables a direct assessment of how convectively generated MSTIDs influence SuperDARN GS observations and trans-ionospheric HF communication channels. This study presents the first results from two-dimensional HF ray-tracing simulations through an MSTID-perturbed ionosphere, modeled using a three-dimensional atmosphere-ionosphere coupled framework. The simulations focus on a severe weather event that occurred over the continental United States (CONUS) in May 2017. Section 2 details the observational data sets, modeling tools, and simulation configurations used in this event study. In Section 3.1, we examine the observations, numerical model outputs, and the resulting data-model comparisons, with a focus on the influence of MSTIDs on Pedersen-mode propagation. Finally, Section 4 summarizes key findings, discusses open challenges, and outlines future research directions in the context of past studies.

2. Data Sets and Methodology

2.1. SuperDARN

Super Dual Auroral Radar Network (SuperDARN) observations are the primary data set used in this study to investigate the impacts of GW-generated MSTIDs on HF propagation. SuperDARN is an international

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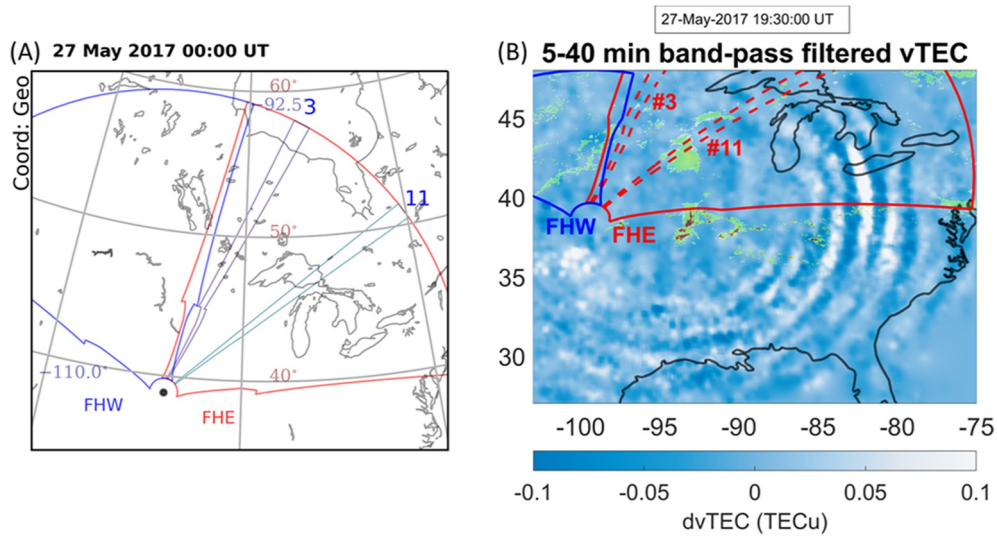


Figure 1. (a) Field-of-views (FoVs) of the SuperDARN Fort Hays East (in blue) and West (in red) radars (up to 50 range gates) located in western Kansas. Beams 3, 11, and 19 of FHE radar are indicated by the colored contours and beam numbers mentioned at the end of the FoV in blue. (b) FoV of the radars overlaid on top of the gridded and interpolated differential Total Electron Content map at 19:30 UT, where the NEXRAD precipitation rate data set is overlaid to highlight the location of the convective source. Concentric traveling ionospheric disturbances propagating from the source are evident, seen as the white and blue alternating wave fronts representing regions of increased and decreased electron densities.

collaboration of HF radars distributed across the middle, high, and polar latitudes of both hemispheres (Chisham et al., 2007; Greenwald et al., 1985; Nishitani et al., 2019). The radars generally operate between 8 and 18 MHz. In normal mode, each radar observes the line-of-sight (LoS) component of plasma velocity along 16 to 20 beams in 75–110 range gates spaced 45 km apart beginning at 180 km range. The typical integration time of each beam sounding is 3 or 6-s, which results in a full radar sweep through all beams in 1 or 2-min, respectively. Figure 1a shows the location of the SuperDARN Fort Hays radars and their Field-of-Views (FoVs, color-coded). Observations from these radars are used in this study. Observations from the SuperDARN Fort Hays East (FHE) radar are primarily used in this study for event characterization and for performing data–model comparisons, while observations from the Fort Hays West (FHW) radar are presented in the Supporting Information S1 to demonstrate the consistent features of the MSTID event under consideration. The primary reason for using the FHE radar observations as the main data source is that the modeling framework (described in Section 2.3) is configured over a computational domain that covers the majority the FHE FoV. Due to computational resource limitations, we restricted the simulations to this model domain. Figure 1b shows the FoVs of the same radars overlaid on differential Total Electron Content (dTEC) observations, which represent spatial perturbations in GNSS-derived TEC relative to a background ionospheric state. The dTEC values are color-coded according to the color bar shown at the bottom of the panel (b). Also shown are precipitation rate measurements from the Next Generation Weather Radar (NEXRAD) network (Crum et al., 1993; Zhang & Gourley, 2018), plotted in dark red/green, indicating the location of thunderstorm activity relative to the radar FoVs.

In this study, a SuperDARN fitted parameter, specifically, backscattered power from the Auto-Correlation Function (ACF (E. Thomas et al., 2025)), is used to understand the impacts of MSTIDs on HF propagation. SuperDARN observations of backscatter are primarily due to two types of targets, ionospheric irregularities and the surface of the Earth, referred to as ionospheric and ground/sea scatters, respectively. Figures 2a and 2b illustrates the different propagation modes, scatter types, and provides an example of a backscattered power FoV plot from the SuperDARN Fort Hays East radar. All SuperDARN observations projected in FoV plots and the corresponding radar FoV contours are mapped in slant range referenced to the ionospheric reflection points associated with ground or sea scatter. The horizontal locations of the GS echoes are determined using the Chisham virtual height model (Chisham et al., 2008). More precise geolocation can be obtained through two-parameter virtual height models that utilize both slant range and elevation angle derived from interferometry (e.g., E. G. Thomas & Shepherd, 2022a; Greenwald et al., 2017). However, elevation angle data were not available for FHE/FHW during this event, and thus the Chisham virtual height model was employed. Based on this model, the

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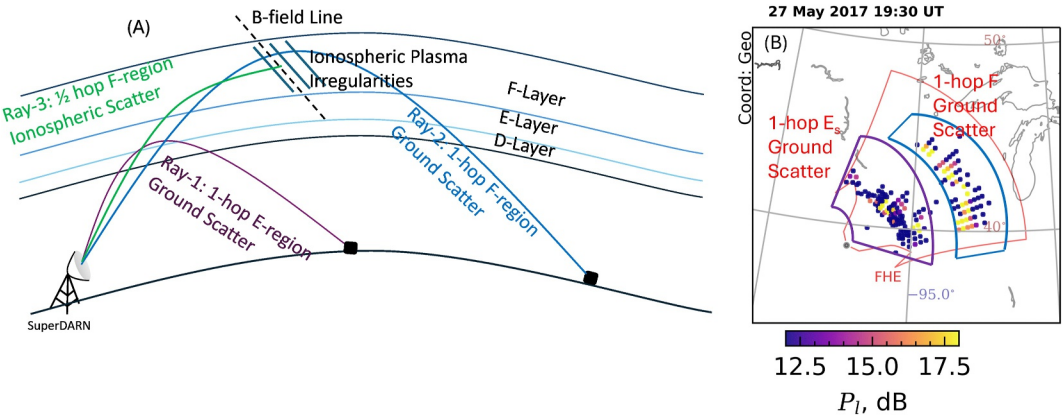


Figure 2. (a) Schematic representation of SuperDARN radar ray paths illustrating ground/sea scatter (GS) and ionospheric scatter. Illustration of 1/2-hop F-region, 1-hop E-region (specifically Es in this case), and 1-hop F-region propagation geometries. The black squares indicate ground backscatter locations. (b) Field-of-view scan from the SuperDARN Fort Hayes East (FHE) radar, displaying line-of-sight backscatter returned power measurements taken on 27 May 2017, at 19:30 UT. The power is color-coded based on the scale provided at the bottom. Various scatter hops are identified and labeled within enclosed regions with red text. All SuperDARN observations and the corresponding radar fields of view are mapped in slant range, referenced to the ionospheric reflection points associated with GS.

median estimated virtual reflection heights are approximately 130 and 350 km for the near-range and far-range scatters, respectively, suggesting that these correspond to E-region and F-region GS. Note that the virtual heights are greater than the true reflection heights; for reference, the h′E and hmF2 data from the Boulder ionosonde presented in the Supporting Information S1 indicate typical E-region and F-region heights of approximately 110–130 km and 230–250 km, respectively, consistent with this interpretation. In this geometry, HF waves transmitted by the radar refract in the ionosphere and return to the radar after reflection from the Earth's surface, producing GS echoes whose slant ranges correspond to the effective propagation path length of the signal (Bristow & Greenwald, 1994).

In the case of ground/sea scatter (GS), due to a high daytime vertical gradient in the ionospheric refractive index, the rays bend toward the ground and are then reflected from surface roughness, before returning to the radar following the same paths. Based on the reflection height and return hops, we can further categorize the GS. Rays 1 and 2 in Figure 2a provide illustrative examples of daytime GS, where Ray 1 and 2 represent rays reflected from E- and F-region altitudes, respectively. Figure 2b presents an example FoV fan plot showing backscattered power from daytime 1-hop E- and F-region ground/sea scatters recorded by SuperDARN Fort Hays East radar. Along with backscatter power, GS "skip distance" can also provide valuable information and insight into the MSTIDs. The "skip distance," defined as the first slant range where GS is detected, is a key parameter that is considerably affected by ionospheric MSTIDs (e.g., Frissell et al., 2014). Ionospheric scatter (IS) occurs when a transmitted signal is reflected from ionospheric irregularities, leading to $\frac{1}{2}$-hop modes (i.e., $\frac{1}{2}$, $1\frac{1}{2}$, $2\frac{1}{2}$; refer to Ray 3 in Figure 2a). Previous studies used SuperDARN back-scattered return power of the GS observations to identify and characterize MSTIDs (e.g., Frissell et al., 2014; Frissell et al., 2016). In this study, we use the backscattered return power of the GS observations to investigate GW-generated MSTIDs over CONUS.

2.2. Global Navigation System Satellites (GNSS) Total Electron Content (TEC)

To corroborate the presence of TIDs over the continental United States (CONUS), Global Navigation Satellite System (GNSS) TEC data were employed as an ancillary data set. The vertical TEC (vTEC) values were derived using a custom-built GNSS data processing pipeline developed by the authors, with methodological details described in Inchin, Deshpande et al. (2025). GNSS phase and pseudorange measurements from GPS, GLONASS, and Galileo satellites were processed for multiple ground stations on 27 May 2017. A fixed ionospheric shell height of 300 km was assumed to determine the ionospheric pierce point coordinates. To extract perturbation signatures indicative of medium-scale TIDs, the vTEC time series were subjected to a high- and band-pass Butterworth filter, isolating fluctuations within the 5–40 min period range.

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2.3. Model Framework

In this section we describe the modeling framework used to estimate the effects of gravity-wave (GW) driven MSTIDs on HF radio propagation. The framework consists of three main components: (a) simulation of atmospheric acoustic–gravity wave generation and propagation using the MAGIC model, (b) simulation of the ionospheric plasma response using the GEMINI model. The overall modeling approach follows the methodology described in detail by Inchin, Heale et al. (2025), where coupled atmosphere–ionosphere simulations were performed for the same severe weather event, and final (c) PHaRLAP HF ray-tracing to geolocate HF rays within the MSTID-perturbed modeled ionosphere.

The generation and propagation of atmospheric GWs were simulated using the three-dimensional nonlinear compressible model MAGIC (Model for Acoustic–Gravity wave Interactions and Coupling) (Snively & Pasko, 2003). In this framework, convective forcing associated with the thunderstorm system is represented through latent heating profiles derived from NEXRAD digital precipitation rate observations (Heale et al., 2019). These heating profiles act as a source term in the energy equation of the MAGIC model and initiate acoustic and GW perturbations that propagate upward from the troposphere into the thermosphere. The background atmospheric state used in the simulations is specified by the empirical NRLMSISE-00 model (Picone et al., 2002), while the background wind fields are constructed by combining MERRA-2 reanalysis winds below 50 km altitude with HWM-14 empirical model winds above this level (Drob et al., 2015). This configuration allows realistic representation of the background atmospheric structure controlling GW propagation.

The resulting MAGIC simulations produce three-dimensional perturbations in neutral density, temperature, and horizontal and vertical velocities that characterize the evolving gravity-wave field. These perturbations propagate upward into the thermosphere where they modify the neutral–plasma coupling and drive TIDs.

To simulate the ionospheric plasma response to these neutral perturbations we use the multi-species nonlinear ionospheric model GEMINI (Zettergren & Semeter, 2012; Zettergren & Snively, 2015). The GEMINI model solves the coupled continuity, momentum, and energy equations for multiple ion species along geomagnetic field lines, enabling self-consistent calculation of ionospheric plasma transport and electrodynamics. In this study, GEMINI is implemented on an open tilted dipole grid with a spatial resolution of approximately 10 km, which provides sufficient resolution to capture MSTID-scale ionospheric structures.

Three-dimensional outputs from the MAGIC simulation—including perturbations in neutral density (O_2, O, N_2), temperature, and fluid velocities—are interpolated onto the GEMINI grid and used as time-dependent drivers with a temporal cadence of 60 s. The background thermospheric state and winds in GEMINI are specified using the same NRLMSIS-00 and HWM-14 models as used in the MAGIC simulations to maintain consistency between the neutral and plasma environments.

To isolate the effects of GWs on the ionosphere, we additionally perform a control GEMINI simulation in which the GW perturbations are excluded while the same background atmospheric state is retained. The difference between the GW-driven and control simulations allows us to quantify the ionospheric perturbations associated with GW-driven MSTIDs.

Finally, the simulated three-dimensional electron density fields from GEMINI are used as inputs for the PHaRLAP HF ray-tracing model to investigate the propagation characteristics of HF radio waves through the perturbed ionosphere. This step enables assessment of how MSTID-induced electron density structures modify HF propagation paths and GS characteristics observed by SuperDARN radars.

2.3.1. PHaRLAP: HF Ray-Tracing Model

We used Provision of High-Frequency Raytracing Laboratory for Propagation Studies (PHaRLAP) ray tracing model to geolocate the SuperDARN rays in the MSTID modified ionosphere (Cervera & Harris, 2014). PHaRLAP incorporates a range of ray tracing engines with varying levels of sophistication, from 2D ray tracing to comprehensive 3D magnetoionic ray tracing. The 2D and 3D ray tracing modules are implementations of the equations developed by C. Coleman (1997); C. J. Coleman (1998) and Haselgrove (1963), respectively. The 2D ray tracing module utilizes ionospheric parameters, HF ray properties, and the elevation and bearing angles of the ray as inputs, generating the height and ground range of the traveling HF ray in kilometers as output. In contrast, the 3D ray tracing module outputs the height, latitude, and longitude of the traveling HF ray. For our research, we

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employed the 2D ray tracing module to geolocate rays along each individual beam of the SuperDARN Fort Hays East (FHE) radar.

2.3.2. Estimating Modeled Backscattered Power

In this study, we were primarily focused on analyzing and comparing SuperDARN GS observations. Output from PHaRLAP was used to estimate the backscattered power.

To estimate the modeled ground/sea scattered power, we adopted the methodology outlined in (De Larquier, 2013). First, simulated rays that reach the ground after reflecting in the ionosphere were identified as GS. These rays are then grouped into 45 km slant-range bins, and the number of rays in each bin is counted. The counts are subsequently weighted by a factor of $\frac{1}{r^3}$, where r is the slant range. This factor accounts for the geometric power decay, assuming the target's size increases linearly with range, as is typical for GS. The choice of 45 km range bins is drawn from the SuperDARN operational range gates. Note that this model does not include the terrain geometry or the reflective properties of the target, which may be separately investigated in the future.

3. Event Analysis: 27 May 2017

In this section, we present a GW-generated ionospheric MSTID event stemming from tropospheric thunderstorm-driven convective source events on 27 May 2017. On 27 May 2017, a severe tropospheric weather outbreak involving a cluster of thunderstorms developed and propagated from Kansas toward Missouri. The associated convective activity generated AGWs that propagated upward into the upper atmosphere and produced MSTIDs observable in TEC measurements (see Figure 1b). These MSTIDs were also observed by the mid-latitude Fort Hays radars (located in central Kansas). We then describe the output of simulated rays from the model framework and compare it with radar observations. We also analyze the influence of MSTIDs on special long ducting HF mode, also referred to as Pedersen mode.

Figure 3 illustrates the impact of MSTIDs on observations from the SuperDARN HF radars. The figure showcases radar FoV scan plots from the SuperDARN Fort Hays radars, strategically located in central Kansas and have FoVs covering a significant portion of central America. Each panel in Figure 3 displays the backscattered fitted power (P_l) obtained from a 1-min radar scan, with the panels presented at 10-min intervals to highlight temporal variations. The backscattered power is color-coded according to the color bar on the bottom left corner. The figure contains six panels, labeled (A) through (F), each displaying a map of backscatter power from FHE/W radars. The time of each scan is shown at the top of each panel, beginning at 18:40 UT and ending at 19:30 UT, with each scan separated by 10 min. Observations reveal how the MSTIDs, driven by convective atmospheric sources, modulate ionospheric conditions, affecting the radar backscatter patterns, specifically skip distances. We see two populations of backscatters within the FoV of FHE/W radars, near (within 30 range gates, $\leq$1,300 km) and far ($\geq$1,500 km) from the radar station. These near-range backscatter signals are most likely produced by sporadic E (Es) layers (e.g., Ogawa et al., 2009; Yakymenko et al., 2015). However, the Es layer observed in this case did not exhibit the characteristics of a blanketing Es layer Kagan et al. (2002), which would otherwise prevent the propagation of HF radio waves to higher altitudes. Blanketing Es layers, as described by Kagan et al. (2002), are dense, thin layers of enhanced ionization in the E-region (typically at 90–120 km altitude) that can completely reflect or absorb high-frequency radio waves, thereby preventing their penetration into the F-region. In contrast, the Es layer in this study allowed partial transmission of HF signals, making it possible to detect F-region backscatter at relatively longer slant ranges. Hence, the near-range scatters are grayed out and are not of interest for this study. The backscatter population far from the radar station is GS. The black and magenta curved lines mark the reference and skip distance of the backscatter population at far distances. The skip distance of the GS population recorded a significant motion (motion toward and away from the radar station) over the 3-hr window. Additionally, we see variations in intensification and lessening of backscatter power values across the beams and with time, indicating ionospheric density compression and rarefactions caused by MSTIDs passing through the FoVs of the radars (Frissell et al., 2014).

We use a different data plotting format, namely range-time-intensity (RTI) plot, to observe motion in skip distance locations along beams 3 and 11 of the FHE SuperDARN radar. Figure 4 presents backscatter power (P_l) along beams 3 and 11 of the FHE radar, as a function of time (horizontal axis) and slant range (vertical axis). The backscatter power is color-coded by color bar on the right of panel (a). The geographic region along these two

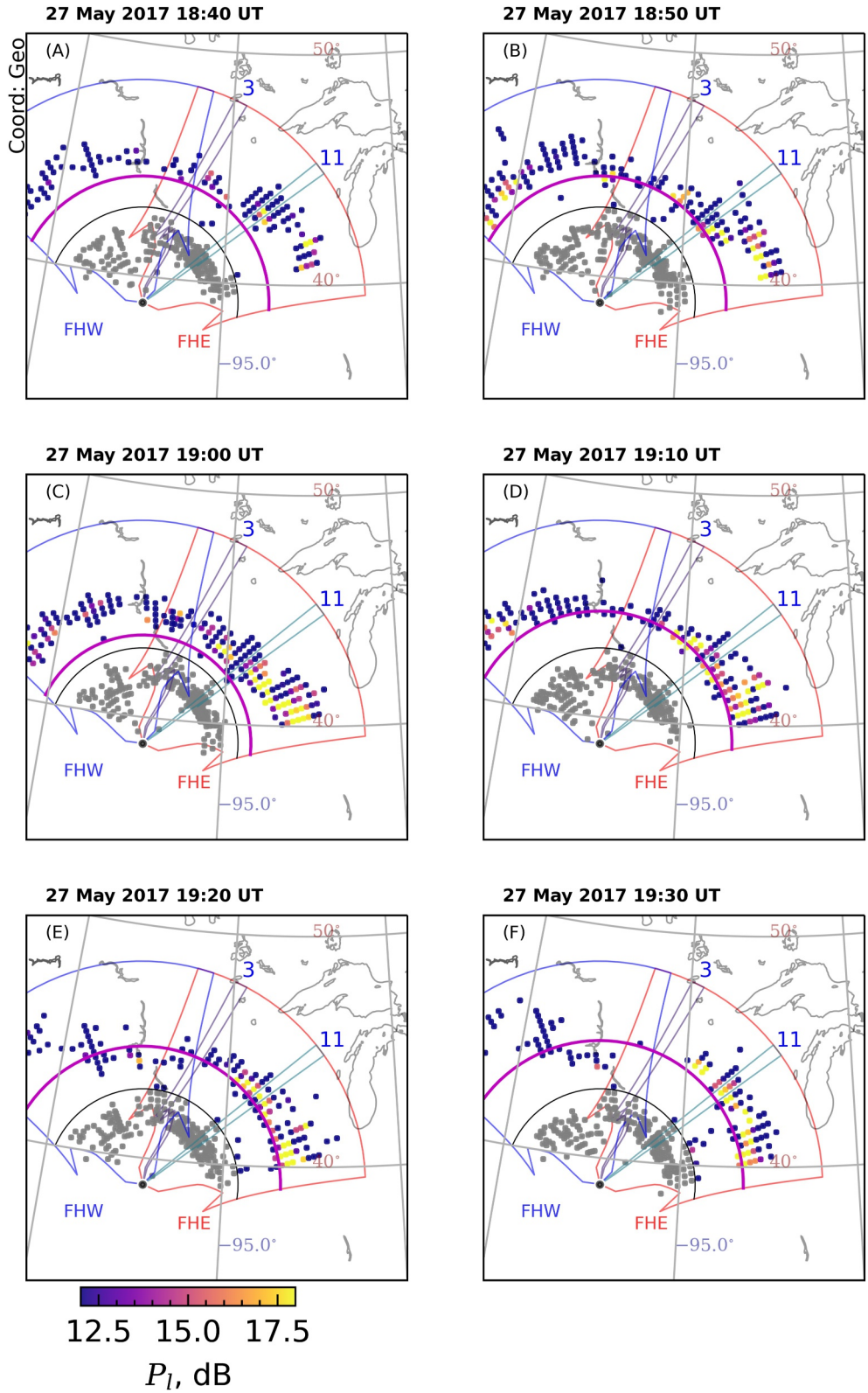


Figure 3.

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beams can be referred back to Figure 3. We see different backscatter populations located at different times and ranges, indicating changes in the ionospheric propagation conditions. We are specifically interested in GS populations between slant range 1,500–2,000 km and 16–21 UT. We have added the green curves in both panels, indicating the motion in skip distance of the GS population located between 1,500 and 2,500 km slant ranges. We see a few notable features in this RTI plot: (a) skip distance of the GS population recorded a significant motion; (b) variations in intensification and lessening of backscatter power values indicating ionospheric density compression and rarefactions. We note that the rarefaction effects are more intense along beam 3 than along beam 11, creating total loss-of-signal between the three different GS populations indicated by green curves. The MSTID activity is quite complex; multiple MSTID wave trains can be seen in the RTI plots by observing changes in skip distance location.

To provide additional context, we have included the FHW RTI plots and corresponding FoV scan maps in the (refer to Figure S1 in Supporting Information S1). The FHW observations shows MSTID structures and its effects HF propagation characteristics. This consistency across paired SuperDARN radars strengthens the robustness of the identified ionospheric disturbance and supports our overall interpretation of the event. A six-panel figure illustrating the spatiotemporal evolution of the thunderstorm system and its ionospheric response is provided in the (Figure S2 in Supporting Information S1). The figure combines dTEC perturbation maps, NEXRAD precipitation observations, and SuperDARN radar measurements to show the progression of the convective system and the associated ionospheric disturbances. The observations reveal the emergence of semi-concentric MSTID structures propagating away from the convective source region, consistent with the upward propagation of thunderstorm-generated atmospheric GWs as noted by Inchin, Heale et al. (2025). Notably, the spatial distribution of SuperDARN GS echoes and the temporal modulation of the estimated skip distance vary coherently with the MSTID phase fronts observed in the dTEC maps, providing additional evidence of the coupling between the convective source and the ionospheric response. This event has been previously studied and modeled using coupled atmosphere–ionosphere frameworks, which demonstrate how deep convective activity generates GWs that subsequently drive MSTID formation (Bergsson et al., 2025; Inchin, Heale et al., 2025).

3.1. Data-Model Comparison

We simulated atmosphere-ionosphere coupled processes related to the 27 May 2017, severe weather system (WS) episode over midwestern United States. This event was chosen due to a geomagnetically calm ionosphere on the day of interest and prior to this date, but also a small presence of TIDs from other sources, whereas TIDs from the WS had distinctive concentric patterns as observed in GNSS TEC (also refer to Figure S1 in Supporting Information S1). Figure 5 illustrates the combined observations from TEC and SuperDARN GS power to investigate the ionospheric signatures of MSTIDs. The figure presents RTI plots of backscatter power from the FHE radar along beams 11 and 19, with the radar slant range mapped to the estimated 1-hop ionospheric reflection points in latitude–longitude coordinates using a virtual height model (e.g., Chisham et al., 2007; E. G. Thomas & Shepherd, 2022b). Detrended vertical TEC (vTEC) perturbations, filtered in the 5–40 min band-pass range to isolate MSTID-scale structures, are overlaid to facilitate comparison with the radar observations. This representation enables a more consistent spatial comparison between the GNSS TEC and SuperDARN data sets; however, a direct one-to-one correspondence between the observed features is not expected. This is primarily due to differences in measurement geometry and sensitivity. GNSS TEC represents a line-of-sight integrated measurement through the full ionosphere, including contributions from the topside F-region, whereas SuperDARN GS observations are predominantly sensitive to bottomside F-region propagation conditions along oblique paths. As a result, differences in peak timing, apparent number of structures, and latitudinal extent are expected between the two data sets.

Despite these differences, both observations consistently indicate the presence of MSTID activity during the interval between approximately 17:00–20:00 UT. In particular, the temporal evolution of TEC perturbations and the modulation of radar backscatter power along the selected beams suggest the passage of large-scale

Figure 3. Fields-of-View (FoV) scan plots from the SuperDARN Fort Hays East (FHE) and West (FHW) radars during GW-generated medium-scale traveling ionospheric disturbances on 27 May 2017: Panels (a–f) present a series of FoV scan plots depicting backscattered power, color-coded according to the scale shown in the bottom-left corner. Beams 3 and 11 of the FHE radar are highlighted with colored contours within the larger FoV outlines and are also labeled in blue text. The magenta solid curve overlaid on the FHE FoV in each panel marks the estimated skip distance of the nearest ground/sea scatter (relevant to this study) returns. The black solid curve serves as a visual reference, a fixed range against which to view the movement of the skip distance. All SuperDARN observations and the corresponding radar FoVs are mapped in slant range, referenced to the ionospheric reflection points associated with ground or sea scatter.

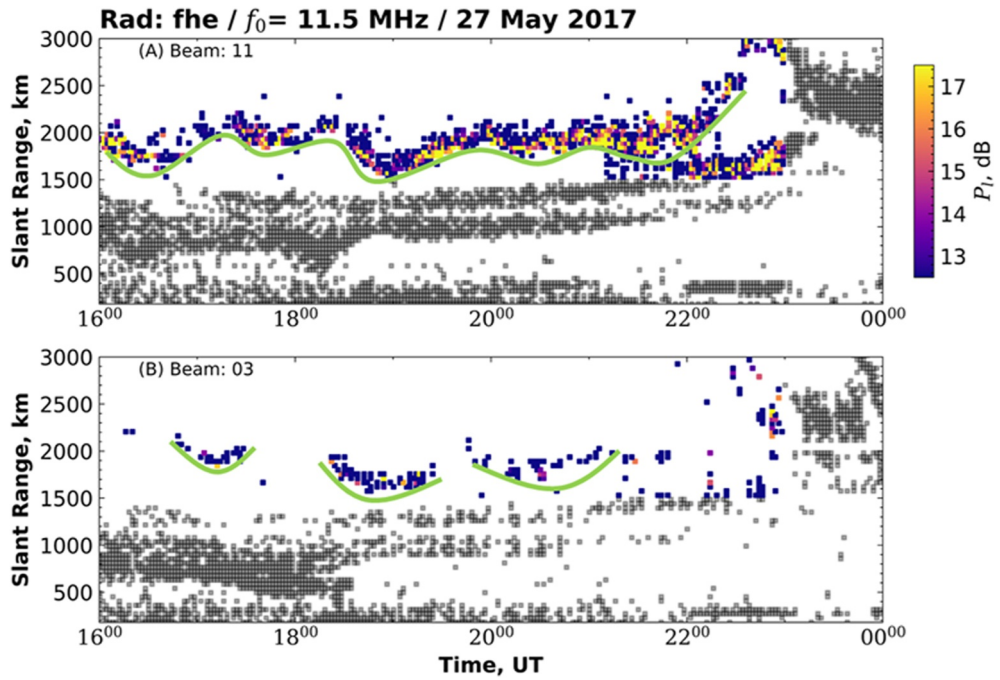


Figure 4. Observations from the SuperDARN Fort Hays East radar along beams (a) 11 and (b) 03: Range-Time-Intensity (RTI) plots of the backscattered power showing HF response to ionospheric medium-scale traveling ionospheric disturbances (MSTIDs) on 27 May 2017, color coded by the color bar on the right. The green curves in both panels indicate the tentative location of the skip distance of the furthest ground/sea scatter population, indicating the medium-scale traveling ionospheric disturbances impacts on the transitionospheric HF communication. The grayed scatter are not of interest for this study.

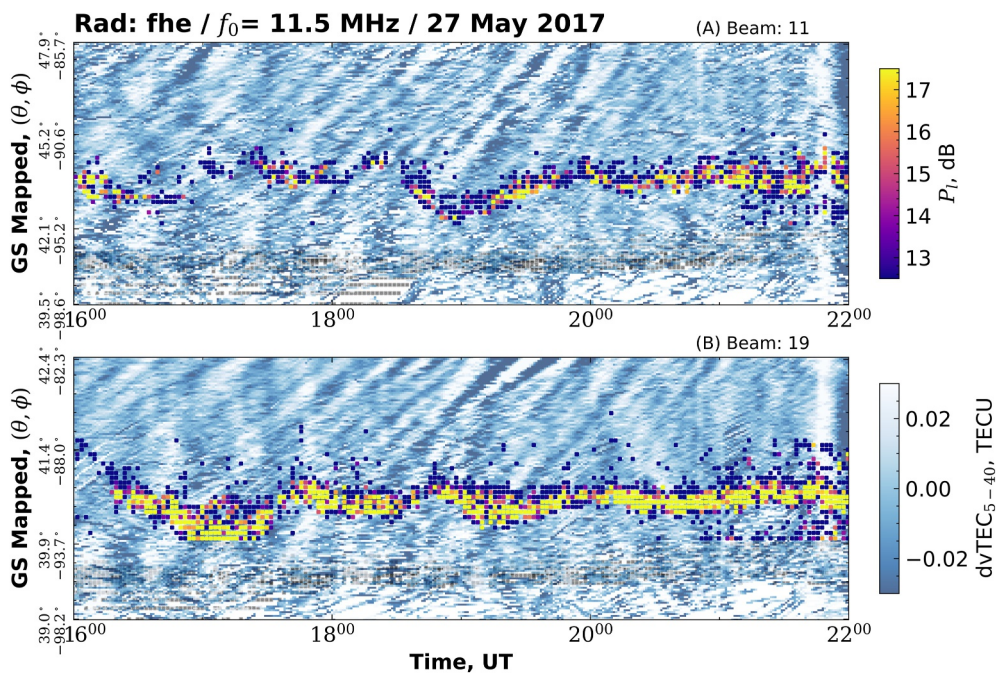


Figure 5. Range–Time–Intensity (RTI) plots showing backscatter power observed by the FHE radar along beams (a) 11 and (b) 19. The range is mapped to the estimated 1-hop ionospheric reflection point in latitude, longitude coordinate. Detrended vertical Total Electron Content perturbations, filtered in the 5–40 min band-pass range, are overlaid for comparison. The radar power observations are color-coded according to the color bars shown on the left panels.

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ionospheric disturbances through the radar field of view. The comparison therefore emphasizes the overall spatiotemporal evolution and characteristic scales of the disturbances rather than exact feature-by-feature agreement, highlighting the complementary nature of GNSS and HF radar observations in capturing MSTID dynamics. As the most prominent TIDs were observed between 17 and 20 UT on 27 May 2017, we constrained our simulations to the 16–22 UT time window to accommodate available computational resources. The event occurred on a geomagnetically calm day, with average daily Dst, Ap, and F10.7 indices of 7 nT, 13 nT, and 84.1, respectively (retrieved from NASA OmniWeb; (King & Papitashvili, 2005)).

Figure 6 shows output from PHaRLAP raytrace through MAGIC-GEMINI simulated ionosphere along the beam 11 of FHE radar (refer to Figure 1 to check the geographic location), which was operating at 11.5 MHz. Panels (a–d) present outputs from four different timestamps, 17:30 UT, 18:15 UT, 19 UT, and 19:30 UT, respectively. Elevation angle of the simulated rays presented in this figure ranges from 18° to 30°. Simulation suggested possible escaping rays (in red), long ducting Pedersen mode rays (in green), and rays reaching the ground (ground/sea scatters), respectively. The black squares on the ground indicate the reflection point of the modeled GS. We overlaid the simulated ionospheric plasma density from MAGIC-GEMINI, and color-coded it by the color bar on the bottom left corner. The dashed black curve indicates the hmF_2 location. The gray shaded region shows the Earth and its curvature (by scale). Contours indicated by black dots represent the group range between 500 and 2,000 km with a separation of 250 km. The vertical line at 1,400 km on ground serves as a reference line to illustrate the back-and-forth motion of the radar skip distance in response to the phase fronts of the MSTID. Simulated MSTID can be observed in the field of view of the radar beam around panel (c–d), which alters the skip distance of the GS rays (the vertical dashed line as reference). We see significant fluctuations in the higher group ranges in response to MSTID. We see that MSTID altered the propagation condition of the ionosphere, creating possibilities of long ducting Pedersen modes at other elevation angles, which is also a function of phase fronts. Simulation shows that the MSTID alters the rays passing through the ionosphere by compression and rarefraction of the plasma density by creating medium-scale (∼100–1,000 km) plasma density structures.

Figure 7 presents the RTI plots showing observed (panel a) and simulated (panel b) GS fitted power (P_l) and relative power (P_r), respectively, from FHE radar along beam 11. Section 2.2.2 describes the methodology to compute P_r from PHaRLAP simulated HF rays. Note that the relative power is not an exact representation of the backscatter power received by the SuperDARN radar, hence, we will not be able to conduct regression error analysis between the observed and simulated power. We, however, think it is possible to conduct a feature-based comparison between observations and simulated outputs. To further quantify the agreement between the observations and the model results, we extend the analysis to include a comparison of the observed and simulated skip-distance variations. Panels (c) and (d) present the time-series comparison of the data and model-derived skip-distance motions and the corresponding distribution of the residual error between the two data sets. This approach provides a quantitative metric for assessing the consistency between the modeled ionospheric perturbations and the observed HF propagation signatures. Note that the observed skip-distance variations are quantified using the estimated GS leading edge. The resulting oscillations reach amplitudes of approximately $\sim\pm$100–200 km (∼5–10 range gates) along selected FHE radar beams, indicating substantial modulation of the HF propagation path during the event. While the overall agreement between the observations and the model results is not perfect, the modeling framework is nevertheless able to reproduce the fluctuating behavior in the skip-distance associated with the event. In particular, the simulations capture the periodicity of the oscillations observed in the data, indicating that the modeled gravity-wave–driven ionospheric perturbations can produce comparable modulation of the HF propagation path. However, the phase alignment between the modeled and observed variations is not well reproduced, which likely reflects uncertainties in the exact source characteristics of the convective system, background ionospheric conditions, and limitations in the spatial coverage of the model domain.

Figure 8 presents the FoV scan plots of backscattered relative modeled power (P_r) from the FHE radar, with panels shown at 30-min intervals to highlight temporal variations. The backscattered power is color-coded according to the color bar shown in the lower-left corner. The time of each scan is displayed at the top of each panel, beginning at 18:00 UT and ending at 20:30 UT, with each scan separated by 30 min. The black and green curved lines mark the reference range and the estimated skip distance of the modeled backscatter population, respectively. Additionally, the observed skip distance (magenta curve; see Figure 3) is overplotted for comparison. The modeled GS population exhibits motion toward and away from the radar over the ∼3-hr interval shown in the figure. Variations in the modeled backscatter power across beams and with time indicate compression and rarefaction of ionospheric plasma density associated with MSTIDs propagating through the radar FoV.

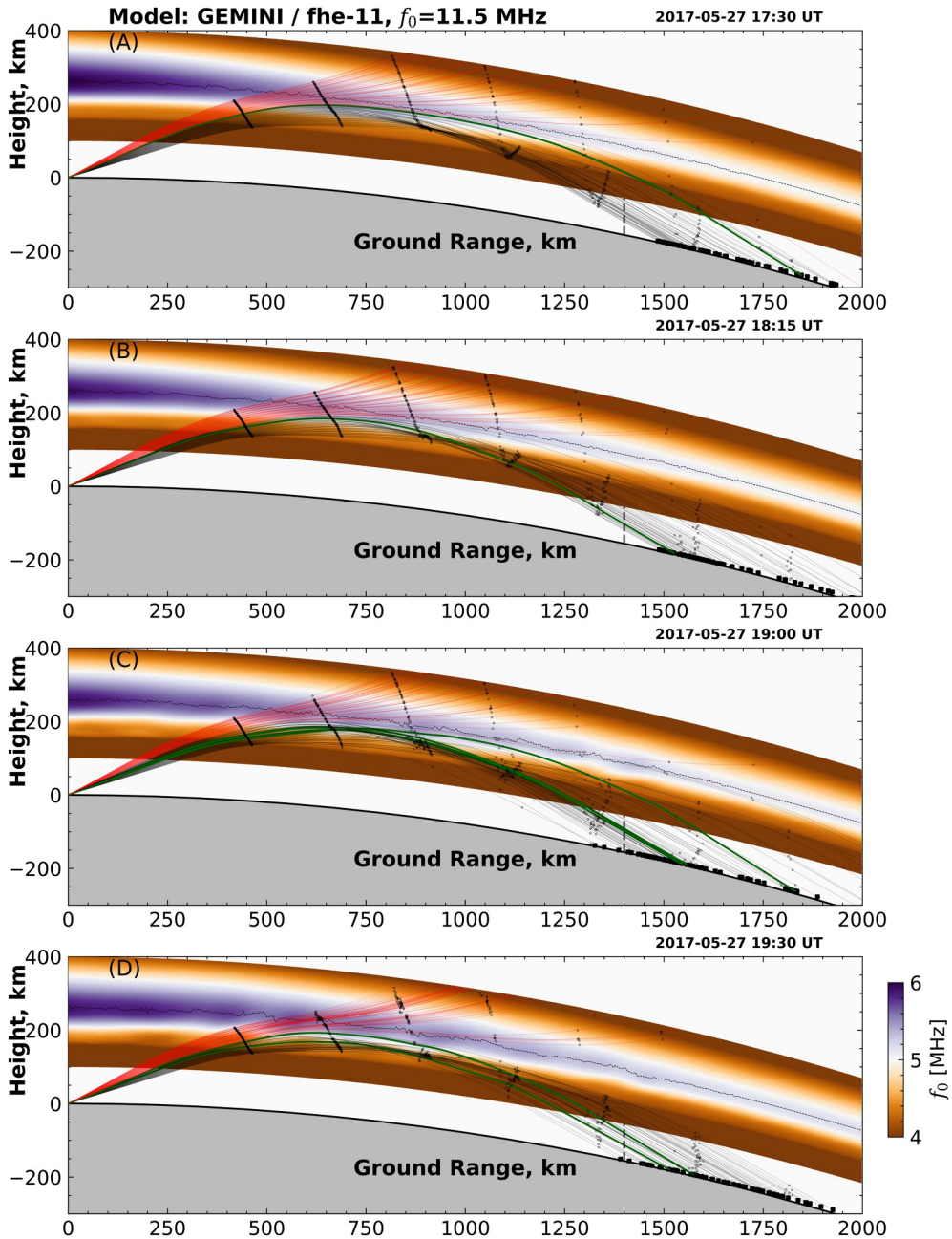


Figure 6. PHaRLAP Raytracing Through MSTID-Modified (MAGIC-GEMINI) Ionosphere on 27 May 2017: Simulated rays from the SuperDARN Fort Hays East (FHE) radar along beam 11 are shown with elevation angles ranging from 18° to 30° at four time instances: (a) 17:30 UT, (b) 18:15 UT, (c) 19:00 UT, and (d) 19:30 UT. The radar was operating at 11.5 MHz. Escaping, long ducting Pedersen mode, and ground rays are indicated by red, thick green, and black curves, respectively. The vertical dashed line at 1,400 km serves as a reference to illustrate the back-and-forth motion of the radar skip distance in response to the phase fronts of the medium-scale traveling ionospheric disturbances. Black squares on the ground indicate the ground/sea scatter reflection points. MAGIC-GEMINI simulated plasma frequencies are overlaid on each panel and color-coded according to the scale shown in the bottom right panel. The dashed black curve indicates the hmF_2. The gray shaded region shows the Earth and its curvature. Black contours indicated by black dots represent the group range between 500 and 2,000 km with a separation of 250 km.

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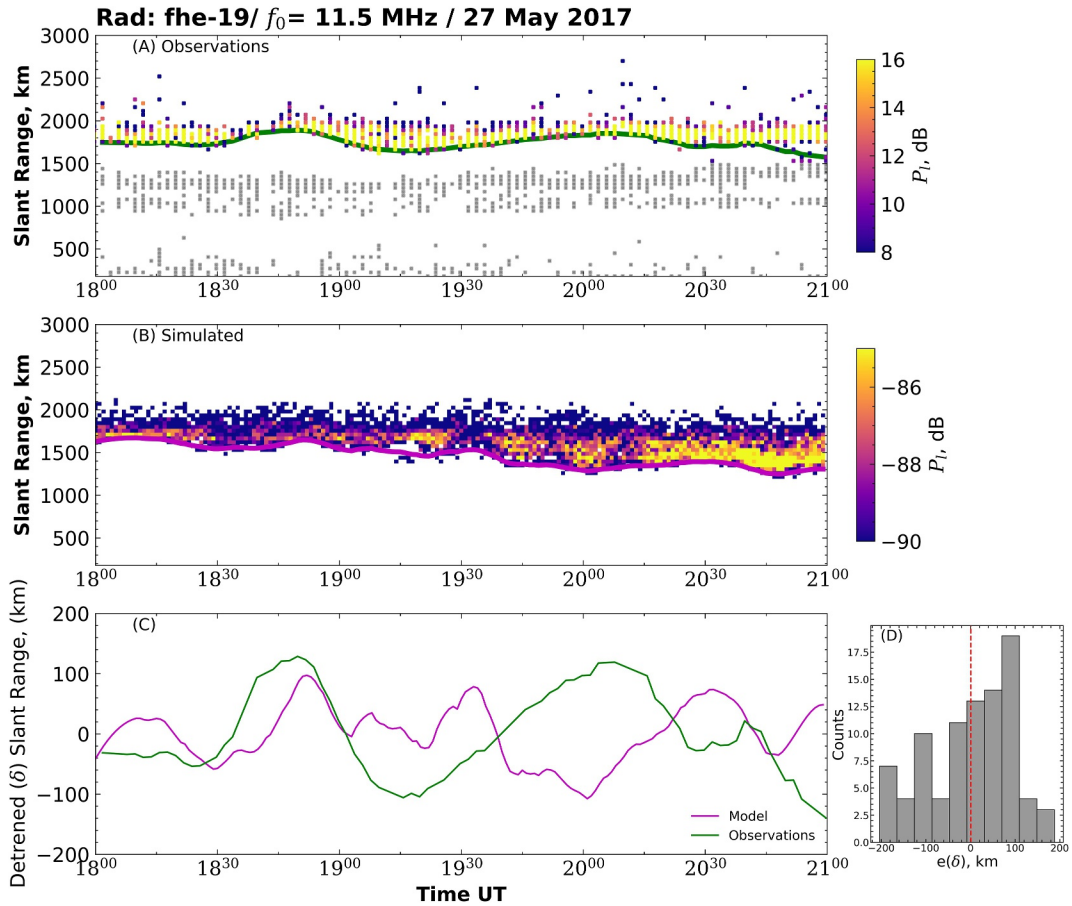


Figure 7. Range-Time-Intensity (RTI) plots are presented, comparing (a) observed and (b) simulated ground/sea scatter (GS) from the SuperDARN FHE radar (beam 19). The black and green curves in the top two panels represent the temporal variations of the observed and modeled skip distances, respectively. Panels (c) and (d) compare the detrended modeled skip distance with the observations and illustrate the corresponding error between the two. These plots illustrate a data-model comparison highlighting the impact of medium-scale traveling ionospheric disturbances (MSTIDs) on the radar-observed GS characteristics. In both panels, observed and simulated power levels are color-coded according to the respective color bars on the right. Backscatter data within the initial 1,500 km slant range are masked (grayed out) as they fall outside the scope of this particular study. The depicted oscillations in GS skip distances, along with concurrent fluctuations in received power, demonstrate the influence of MSTIDs on transionospheric HF radio wave propagation, which directly affects communication channels.

The level of agreement between the observations and the model varies across different time intervals. For example, relatively good correspondence is observed between approximately 18:40 and 19:10 UT, whereas the agreement is weaker at other times. Such discrepancies are expected for several reasons. First, the convective source used to drive the gravity-wave simulations is derived from precipitation observations and therefore represents an approximation of the true thunderstorm dynamics, which introduces uncertainty in the timing and amplitude of the generated waves. Second, background ionospheric conditions in the model are specified using empirical models, which will not fully capture the actual state of the ionosphere during the event, particularly in capturing the physics at smallest scales, and behavior resulting from physical processes not included in the model. Third, the coupled MAGIC–GEMINI framework simulates the generation and propagation of GWs self-consistently, but small differences in wave propagation speed or source timing can produce phase offsets between the modeled and observed MSTID structures. Additional factors, such as the presence of sporadic E layers and limitations in the HF ray-tracing assumptions, may also influence the observed backscatter signatures.

Therefore, rather than expecting a one-to-one correspondence between the observations and the simulations, the objective of this comparison is to determine whether the model reproduces the key characteristics of MSTID-driven HF propagation effects, including the temporal modulation of skip distance and the spatial variability

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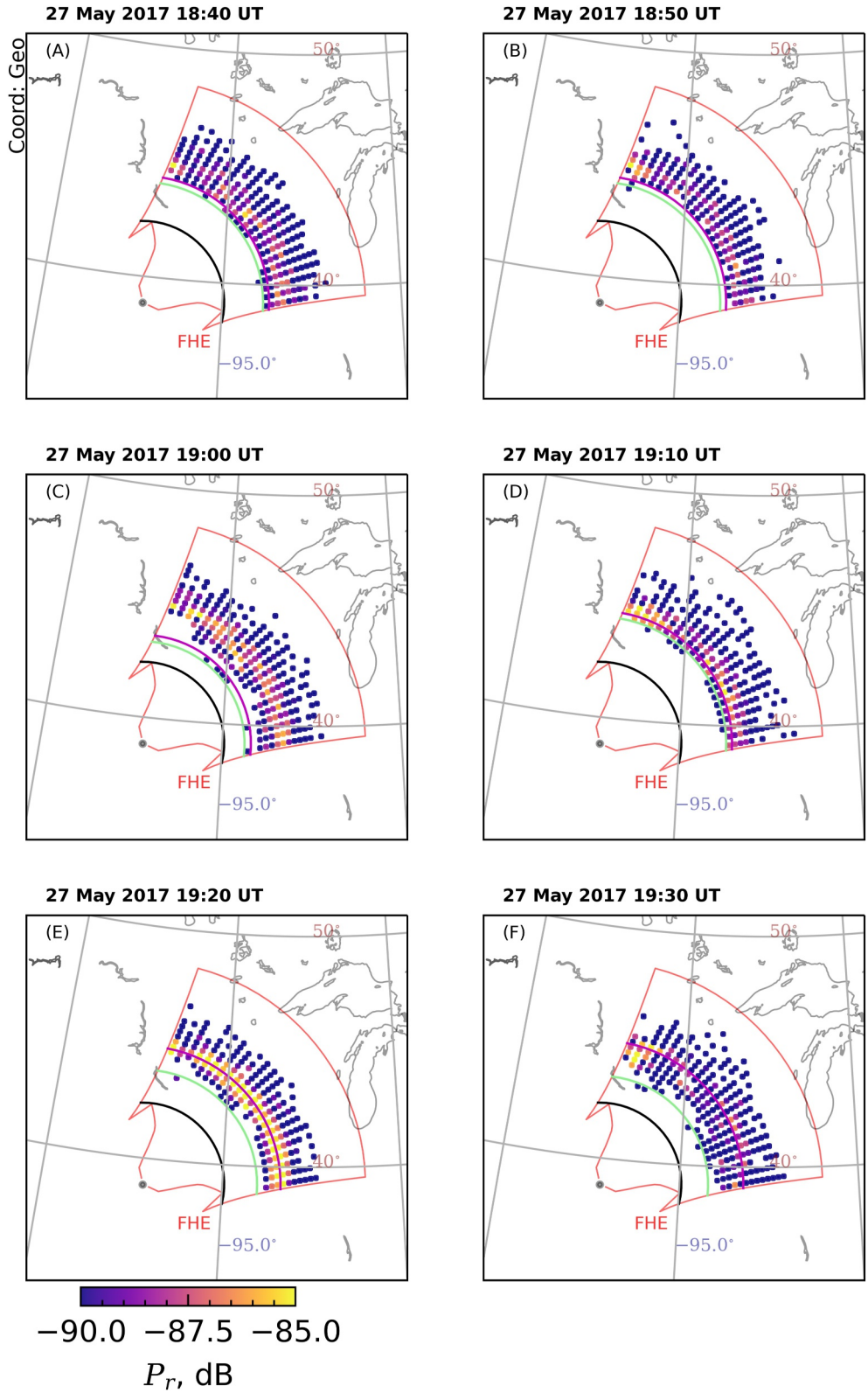


Figure 8.

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in backscatter power. In this respect, the simulations capture the overall fluctuating behavior of the HF propagation paths associated with MSTID activity.

3.2. Special Long Ducting HF Propagation Mode: Pedersen Mode

While MSTID-driven oscillations in ionospheric electron density affect the skip distance by perturbing the trajectory of transionospheric HF rays, our simulations also reveal the potential formation of long-distance ducted propagation modes, commonly referred to as Pedersen modes (Ponomarenko et al., 2015). These modes occur within a narrow angular range of elevation and are characterized by divergent rays that glide along the ionospheric peak (hmF_2) (Ponomarenko et al., 2015). In an unperturbed ionosphere, such Pedersen modes are expected to exist only within a narrow communication channel, determined by specific combinations of elevation angle and frequency. However, our simulations suggest that in a TID-perturbed ionosphere, multiple such narrow channels may emerge, each capable of supporting Pedersen-like ducted propagation (see Figure 6). While no observational evidence of Pedersen modes was detected by the Fort Hays radars in this particular case, the simulations suggest that MSTID-induced perturbations may create conditions favorable for their occurrence.

To investigate this further, we present a focused analysis of modeled Pedersen-mode propagation in MSTID-perturbed conditions using Figure 9. We compare a narrow elevation angle range (21.7°–21.9°) at 11.5 MHz between a TID-perturbed ionosphere (Panel a) and a control simulation (Panel b) on 19:30 UT. Three rays were launched within this range at 0.1° separation. In the control case, no ducting or Pedersen mode is observed. However, in the presence of TIDs, a narrow communication channel emerges around 21.8°, supporting a ducted Pedersen mode. Panel a of Figure 9 shows a zoomed-in view of the three rays, highlighting the interaction between the MSTID structures and ray trajectories. All rays are influenced by the TID phase fronts, as evident from their modified skip distances compared to the control run. Of particular interest is the green-colored ray, which exhibits a gliding trajectory aligned with the ionospheric peak (hmF_2), shown by the black dashed curve. The top-right inset in Panel a illustrates how the TID structure facilitates upward bending and rotation of the ray, enabling it to propagate parallel to the ionospheric maximum over a ground range exceeding 250 km. This behavior indicates that the MSTID crest contributes to localized density refraction, allowing rays at lower elevation angles to reach higher altitudes and transition into Pedersen mode propagation.

We further analyzed the occurrence of Pedersen modes by comparing ray trajectories simulated between elevation angles of 18° and 24° at 19:00 UT, as shown in Figure 10. Panels a and b present the refractive index η, plotted as a function of slant range and elevation angle. These refractive index distributions are derived from GEMINI simulations with and without MSTID (Control). Panel c of Figure 10 shows the refractive index η along the path of a selected ray at an elevation angle of 22.7°, under both MSTID-perturbed (blue) and control (red) conditions. Several key observations emerge from this figure:

- Long ducting modes appear at relatively lower elevation angles (specifically between 21° and 23°), when compared with quiet conditions, in the MSTID simulation.
- Rays in the MSTID case traverse alternating regions of low and high plasma density, creating favorable propagation conditions for long-distance ducting at lower elevation angles.
- Note that higher and lower densities lead to lower and higher refractive indices, respectively. Localized MSTID structures (see Figure 9a) act as waveguides by steering rays through regions of low refractive index toward the ionospheric peak (hmF_2), enabling extended gliding and longer skip distances and creating HF conditions suitable for supporting long-ducting Pedersen mode.
- It is noteworthy that only a discrete subset of elevation angles exhibits the full extent of MSTID-induced variations, most likely corresponding to rays whose propagation paths are resonantly aligned with the MSTID phase fronts.

Figure 8. Fields-of-view (FoV) scan plots of the backscattered from the SuperDARN FHE radar simulated using MAGIC-GEMINI and raytraced using PHaRLAP. Panels (a–f) present a series of FoV scan plots depicting relative backscattered power, color-coded according to the scale shown in the bottom-left corner. The magenta and green solid curves overlaid on the FHE FoV in each panel mark the estimated skip distance of the furthest simulated and observed ground/sea scatter (GS) returns, respectively. The black solid curve serves as a visual reference, indicating the extent of the most distant GS population. All SuperDARN observations and the corresponding radar FoVs are mapped in slant range, referenced to the ionospheric reflection points associated with ground or sea scatter.

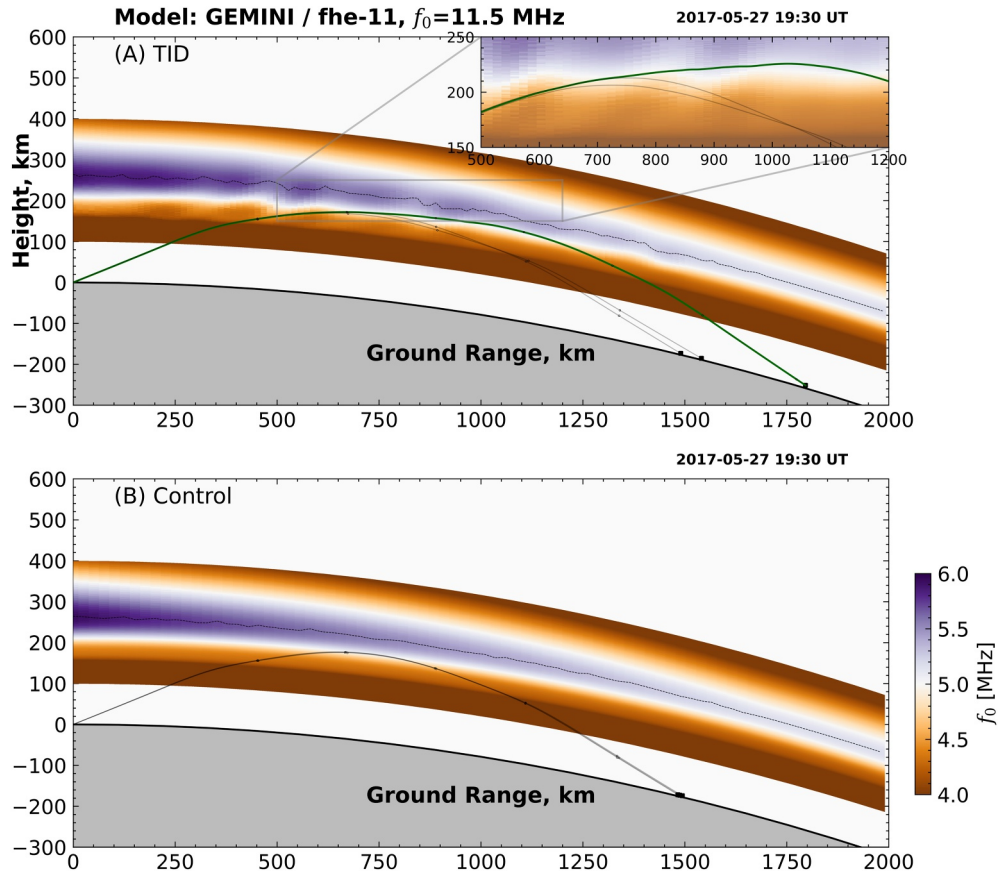


Figure 9. PHaRLAP raytracing through (a) MSTID-modified, and (b) controlled ionosphere showing impact of medium-scale traveling ionospheric disturbances on long-ducting HF modes. Simulated rays from the SuperDARN Fort Hays East (FHE) radar along beam 11 are shown with elevation angles ranging from 21.7° to 21.9° at 19:30 UT. The radar was operating at 11.5 MHz. MAGIC-GEMINI simulated plasma frequencies are overlaid on each panel and color-coded according to the scale shown in the bottom right panel.

4. Discussions and Summary

This study investigates the impact of gravity wave (GW)-driven medium-scale traveling ionospheric disturbances (MSTIDs), generated by a severe thunderstorm over the Midwest and eastern United States on 27 May 2017, on high-frequency (HF) communication channels using SuperDARN radar observations and first-principles modeling. 3D MAGIC model, driven by NEXRAD-derived precipitation rates, was used to compute temporally and spatially varying latent heating profiles as GW sources. These outputs were then used to drive the GEMINI 3D multi-fluid nonlinear ionospheric model to simulate ionospheric plasma responses. The coupled MAGIC-GEMINI results were validated against GNSS vTEC and COSMIC-2 radio occultation observations and presented in a separate study, demonstrating good agreement in MSTID periods (∼5–25 min) and horizontal wavelengths (∼100–300 km). Previous studies showed that the simulated vertical TEC fluctuations (∼0.02–0.35 TECU) matched observations within an uncertainty of ∼15%, suggesting high model accuracy for the event under consideration. These results provide a robust basis for conducting 2D PHaRLAP ray-tracing simulations to examine MSTID impacts on HF propagation observed by SuperDARN radars.

Both observations and model results reveal that GW-driven MSTID significantly affect HF communication channels. Observational data show notable fluctuations in GS skip distances, leading to asymmetric backscattered power distributions across both time and space, as evident in SuperDARN RTI and FoV plots. The modeling framework successfully reproduces MSTID-driven signatures in HF data, demonstrating its utility for studying convective source-driven ionospheric perturbations. The observed spatial asymmetry in signal power across radar beams indicates horizontally structured TID fronts, while temporal asymmetries reflect plasma density fluctuations caused by compression and rarefaction effects. These features are consistent with prior studies on MSTID

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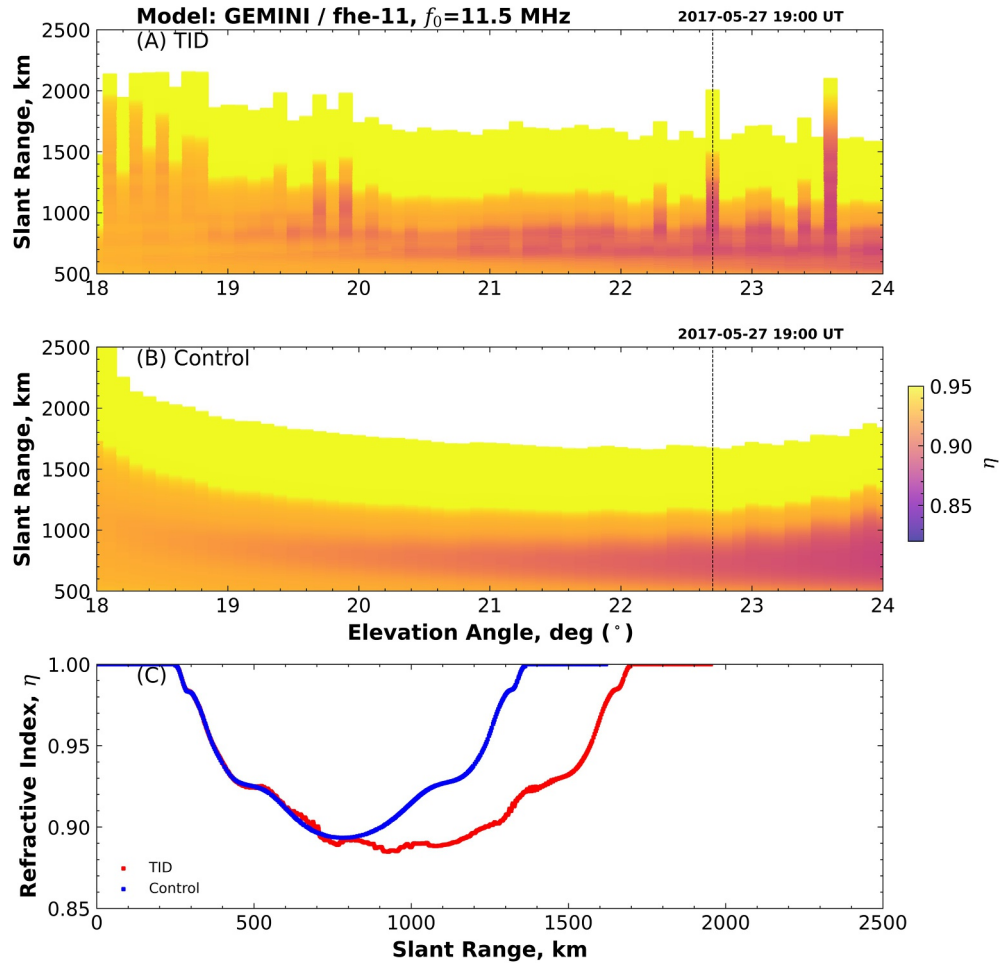


Figure 10. The figure illustrates the simulated distribution of refractive index (η) as a function of slant range and elevation angle at 19:00 UT along beam 11 of the FHE radar operating at 11.5 MHz. Panels (a) and (b) compare propagation through an MSTID-modified ionosphere and a controlled (unperturbed) ionosphere, respectively, thereby demonstrating the impact of medium-scale traveling ionospheric disturbances on long-range HF ducting modes. In both panels (a) and (b), the refractive index (η) is color-coded according to the scale shown by the color bar adjacent to panel (b). Panel (c) displays the refractive index profile as a function of slant range along a specific elevation angle of 22.8°. This panel highlights how MSTID-induced fluctuations in the refractive index can create conditions favorable for the formation of a long-range Pedersen ducting mode.

impacts using HF observations (Frissell et al., 2014, 2022; Prikryl et al., 2022), which predominantly focused on auroral, magnetospheric, or volcanic sources (Kozlovsky et al., 2025; Zhang et al., 2022). In contrast, our study emphasizes that convective tropospheric sources, such as severe thunderstorms, can also generate GWs capable of significantly disrupting HF propagation.

Unlike the wintertime event analyzed by Frissell et al. (2016), which exhibited pronounced periodic slanted traces in GS power associated with equatorward MSTID propagation, the 27 May 2017 event presented here shows comparatively weaker modulation in backscatter power despite clear skip-distance variations. One key difference lies in the seasonal background conditions. The Frissell et al. (2016) case occurred during Northern Hemisphere winter, when F-region GS is typically stronger and more coherent, allowing MSTID-induced ray focusing to produce distinct power enhancements that shift in range with time. In contrast, the present event occurred during summer, when ionospheric conditions are often more structured and variable. Independent observations from the Boulder, CO (BC840) digisonde indicate the presence of sporadic E (Es) during the event (see Figure S3 in Supporting Information S1). Far-range echoes (≥1,700 km slant range) exhibiting MSTID-like features, while near-range gates (between ~500–1,700 km slant range) are dominated by E-region scatter, are also evident in other SuperDARN observations on the event day and adjacent days, suggesting possible Es-related effects.

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Sporadic E layers can modify HF skywave propagation through additional reflection paths and frequency-dependent refraction associated with sharp electron density gradients, potentially introducing multipath propagation and phase distortions in the transmitted and received signals (Cameron et al., 2022; Davies, 1990; Haldoupis, 2011). While the precise contribution of Es to the observed radar signatures cannot be definitively determined in this case, such effects could partially obscure or smear the periodic power oscillations expected from pure F-region MSTID-driven focusing/defocusing effects. Thus, while skip-distance modulation remains evident, the absence of strong, periodic slanted power traces may reflect the combined influence of seasonal background conditions and possible concurrent Es activity.

We explore how the presence of MSTIDs modulates the behavior of Pedersen mode propagation. In the absence of disturbances, as shown in Figure 6a, the propagation characteristics of Pedersen rays are primarily governed by the electron density distribution near the peak of the F-layer. Prior simulation work by Eckermann and Preusse (1999) demonstrated that when the F1 and F2 layers are closely spaced, the resulting flattened density profile minimizes energy leakage, thereby creating favorable conditions for sustained Pedersen mode propagation.

Our modeling results indicate that MSTID-induced ionospheric disturbances can enhance the likelihood of Pedersen modes emerging at lower elevation angles. This suggests that both structured and irregular plasma density features introduced by MSTIDs play a critical role in supporting these long-ducting propagation paths. Consistent with the findings of Eckermann and Preusse (1999), our simulations suggest that the plasma structures near the F-layer peak act as stabilizing features for Pedersen mode propagation during disturbed periods.

Our future studies will focus on evaluating the impacts of MSTID-driven long-ducting Pedersen modes and how MSTID phase fronts (i.e., ionospheric tilts) influence HF communication. To do this, we will employ PHaRLAP's 3D ray-tracing capabilities to examine the three-dimensional structure of ionospheric tilts. Additionally, we aim to assess the effectiveness of existing HF sounding techniques in capturing MSTID characteristics. Although HF sounder observations are limited in both spatial coverage and temporal resolution, model-guided analysis will allow us to evaluate the accuracy of key ionospheric parameters, such as NmF_2 and hmF_2—derived from instruments like Digisonde and VIPIR. This integrated approach will help identify which sounding techniques best capture the dynamic behavior of the ionosphere, especially during MSTID events.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

GNSS signal observations used for the investigation of MSTIDs in the study are available at (a) Scripps Orbit and Permanent Array Center via http://sopac-csrc.ucsd.edu/index.php/gnss/ with public access, (b) The National Aeronautics and Space Administration the Crustal Dynamics Data Information System (Noll, 2010) via https://cddis.nasa.gov/archive/gnss/data/daily/ with public access, (c) National Oceanic and Atmospheric Administration's National Geodetic Survey NGS via https://geodesy.noaa.gov/CORS/ with public access and GAGE Facility operated by UNAVCO, Inc. via https://www.unavco.org/data/gps-gnss/gps-gnss.html with public access. GAGE Facility, operated by UNAVCO, Inc., is supported by the National Science Foundation, the National Aeronautics and Space Administration, and the U.S. Geological Survey under NSF Cooperative Agreement EAR-1724794. All the data and Python code are uploaded into the Zenodo available for public use (Chakraborty, 2025). The analysis and visualization were completed with the help of free, open-source software tools such as matplotlib (Hunter, 2007), IPython (Perez & Granger, 2007), pandas (McKinney, 2010), pyDARN (Shi et al., 2022), and others (e.g., Millman & Aivazis, 2011). The SuperDARN data is available via http://vt.superdarn.org/.

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